Project Executable Files

Date	04 july 2026
Team ID	
Project Name	Credit Card Approval Prediction
Maximum Marks	3 Marks

Project Executable Files
This section captures submission of all executable and deployable files for the Credit Card Approval Prediction System project. All files are organized, named, and uploaded to the specified submission platform so the evaluator can run or deploy the solution using the provided files and instructions.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / package.json / dependency file	Yes
4	Database schema / seed files (if applicable)	NA
5	Environment configuration file (.env.example or similar)	Yes
6	Deployed application URL (if hosted)	Yes
7	APK / Executable binary (if applicable for mobile/desktop apps)	NA
8	Dockerfile / Containerization config (if applicable)	NA
9	Test files and test results	Yes
10	Demo video	Yes

Step 2: File / Folder Structure

Overview of the project's file and folder structure:

```
/credit-card-approval-prediction
  /dataset          - Raw and processed credit card approval dataset (CSV)
  /notebooks        - Jupyter notebooks for EDA, preprocessing, and model training
  /models           - Saved trained models (best_model.pkl, encoders, scalers)
  /static           - CSS, JS, and image assets for the Flask web app
  /templates        - HTML templates (home.html, predict.html, result.html)
  app.py            - Main Flask application (routes and prediction logic)
  watson_deploy.py  - IBM Watson Machine Learning deployment script
  requirements.txt  - Python dependencies
  .env.example      - Sample environment configuration (IBM Watson API keys, etc.)
  README.md         - Setup and run instructions
```

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	https://credit-card-approval-prediction-lgxc.onrender.com

Login Credentials (Demo)	Not required — the application is a public prediction form with no login.
Platform / Hosting Provider	IBM Watson Machine Learning (model hosting) with Flask app deployed on Render
Repository Link	https://github.com/akhilreddy772/CreditCardApprovalPrediction.git
Demo Video Link	https://drive.google.com/file/d/1XdAzNJilsy9IMExG5o2F83zwQ98j0BrY/view?usp=drive_link

Step 4: Run Instructions

Step-by-step instructions for the evaluator to run the project locally:

Pre-requisites:

Python 3.8 or above, pip, Anaconda Navigator (optional), a web browser, and an internet connection for IBM Watson ML access.

Step 1: Clone the repository: git clone https://github.com/TEAM-CCAPS-01/credit-card-approval-prediction.git

Step 2: Install dependencies: pip install -r requirements.txt

Step 3: Configure environment: copy .env.example to .env and add your IBM Watson API key and deployment URL

Step 4: (Optional) Retrain the model: run the notebooks in /notebooks to regenerate best_model.pkl

Step 5: Start the application: python app.py

Step 6: Open browser at http://localhost:5000

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	Prediction accuracy drops slightly for applicants with very limited credit history (thin-file applicants).	Flagged for manual review by the credit analyst; documented as a future improvement area.
2	IBM Watson ML endpoint occasionally has cold-start latency on the first request after inactivity.	Added a loading indicator on the web app; subsequent requests respond in real time.
3	The app currently supports English-language input only.	Out of scope for this version; noted as a potential future enhancement.